Linear Algebra and Applications

Lecture 14-15 Least Squares Method

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Abstract

These lecture notes introduce the mathematical framework for solving the **least squares** problem, which addresses systems of linear equations ($\mathbf{A}\mathbf{x} = \mathbf{b}$) where an exact solution does not exist. The core objective is defined as finding the vector $\hat{\mathbf{x}}$ that minimizes the norm of the **residual vector** ($\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$), thereby finding the closest possible solution. Algebraic and geometric interpretations are provided, along with practical applications in linear regression and data science.

Contents

1 Introduction to Least Squares

1.1 The Fundamental Problem

In many practical situations, we encounter systems of linear equations where the number of equations exceeds the number of unknowns:

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

where:

- $\mathbf{A} \in \mathbb{R}^{m \times n}$ is an $m \times n$ matrix
- $\mathbf{x} \in \mathbb{R}^n$ is the unknown vector
- $\mathbf{b} \in \mathbb{R}^m$ is the observation vector
- Typically $m > n$ (more equations than unknowns)

Definition

The Challenge: When $\mathbf{b} \notin R(\mathbf{A})$ (i.e., $\mathbf{b}$ is not in the column space of $\mathbf{A}$), there is **no exact solution**.

1.2 The Residual Concept

We define the **residual vector** as:

$$\mathbf{r}(\mathbf{x}) = \mathbf{A}\mathbf{x} - \mathbf{b}$$

The magnitude of this residual measures how "good" our solution is. When no exact solution exists, we seek the "best possible" solution.

Definition

Least Squares Objective: Find $\hat{\mathbf{x}}$ that minimizes the squared Euclidean norm of the residual:

$$\min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$$

1.3 Why Squared Norm?

- **Differentiable everywhere** (unlike the absolute value)
- **Emphasizes larger errors** more heavily
- **Leads to nice analytical solutions**
- Equivalent to minimizing $(\mathbf{A}\mathbf{x} - \mathbf{b})^T(\mathbf{A}\mathbf{x} - \mathbf{b})$

2 Geometric Interpretation

2.1 Visualizing the Problem

Consider a simple case with $m = 3$ and $n = 2$:

- The columns of $\mathbf{A}$ span a plane in $\mathbb{R}^3$ (the column space $R(\mathbf{A})$)

- Vector $\mathbf{b}$ is typically not in this plane
- We want to find the point in $R(\mathbf{A})$ closest to $\mathbf{b}$

Geometric interpretation:
$R(\mathbf{A})$ is a subspace (plane through origin)
$\mathbf{b}$ lies outside this plane
Find $\mathbf{A}\hat{\mathbf{x}} \in R(\mathbf{A})$ closest to $\mathbf{b}$

2.2 Orthogonality Principle

Theorem

Theorem (Orthogonality Principle): The optimal residual $\hat{\mathbf{r}} = \mathbf{A}\hat{\mathbf{x}} - \mathbf{b}$ must be **orthogonal** to every vector in $R(\mathbf{A})$.
Mathematically: $\hat{\mathbf{r}} \perp R(\mathbf{A})$ or $\hat{\mathbf{r}} \in R(\mathbf{A})^{\perp}$

Why? If the residual weren't orthogonal, we could find a point in $R(\mathbf{A})$ even closer to $\mathbf{b}$.

3 Derivation of Normal Equations

3.1 From Geometry to Algebra

The orthogonality condition can be expressed mathematically:

$$\mathbf{w}^T(\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}) = 0 \quad \text{for all } \mathbf{w} \in R(\mathbf{A})$$

Since any $\mathbf{w} \in R(\mathbf{A})$ can be written as $\mathbf{A}\mathbf{v}$ for some $\mathbf{v}$, we have:

$$(\mathbf{A}\mathbf{v})^T(\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}) = 0 \quad \text{for all } \mathbf{v}$$

3.2 Step-by-Step Derivation

1. **Start with orthogonality condition:**
$$(\mathbf{A}\mathbf{v})^T(\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}) = 0 \quad \forall \mathbf{v}$$
2. **Apply transpose property:**
$$\mathbf{v}^T\mathbf{A}^T(\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}) = 0 \quad \forall \mathbf{v}$$
3. **For this to hold for all $\mathbf{v}$:**
$$\mathbf{A}^T(\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}) = \mathbf{0}$$
4. **Rearrange to get Normal Equations:**
$$\boxed{\mathbf{A}^T\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^T\mathbf{b}}$$

3.3 Properties of Normal Equations

- **Size:** The system $\mathbf{A}^T\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^T\mathbf{b}$ is $n \times n$ (smaller than original $m \times n$)
- **Symmetry:** $\mathbf{A}^T\mathbf{A}$ is always symmetric
- **Positive Semi-definite:** $\mathbf{A}^T\mathbf{A}$ is always positive semi-definite
- **Always solvable:** $\mathbf{A}^T\mathbf{b}$ is always in the column space of $\mathbf{A}^T\mathbf{A}$

4 Existence and Uniqueness of Solutions

4.1 Always Existence

Key Concept

Key Fact: The normal equations $\mathbf{A}^T\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^T\mathbf{b}$ **always have at least one solution**, regardless of $\mathbf{A}$.

Why? Because $\mathbf{A}^T\mathbf{b}$ is always in the column space of $\mathbf{A}^T\mathbf{A}$.

4.2 Uniqueness Condition

The uniqueness of the solution depends on the **rank** of matrix $\mathbf{A}$:

4.2.1 Case 1: Unique Solution (Full Column Rank)

If $\mathbf{A}$ has **linearly independent columns** (rank $= n$), then:

- $\mathbf{A}^T\mathbf{A}$ is **invertible**
- The unique solution is:
$$\boxed{\hat{\mathbf{x}} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{b}}$$
- The matrix $(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$ is called the **Moore-Penrose pseudo-inverse** of $\mathbf{A}$

4.2.2 Case 2: Infinite Solutions (Rank Deficient)

If $\mathbf{A}$ has **linearly dependent columns** (rank $< n$), then:

- $\mathbf{A}^T\mathbf{A}$ is **singular** (not invertible)
- There are infinitely many solutions
- All solutions give the **same minimum residual**
- The general solution is:
$$\hat{\mathbf{x}} = \hat{\mathbf{x}}_p + \mathbf{w}, \quad \mathbf{w} \in N(\mathbf{A})$$
where $\hat{\mathbf{x}}_p$ is any particular solution and $N(\mathbf{A})$ is the null space of $\mathbf{A}$

4.3 Proof of Optimality (Pythagorean Theorem)

Theorem

Theorem (Optimality): $\hat{\mathbf{x}}$ minimizes $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$ **if and only if** it satisfies $\mathbf{A}^T\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^T\mathbf{b}$.

Proof: Let $\hat{\mathbf{x}}$ satisfy the normal equations, and let $\mathbf{x}$ be any other vector.

1. Compute the residual for $\mathbf{x}$:
$$\mathbf{r} = \mathbf{A}\mathbf{x} - \mathbf{b}$$
2. Write $\mathbf{r}$ in terms of $\hat{\mathbf{r}}$:
$$\mathbf{r} = (\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}) + \mathbf{A}(\mathbf{x} - \hat{\mathbf{x}}) = \hat{\mathbf{r}} + \mathbf{A}(\mathbf{x} - \hat{\mathbf{x}})$$

3. Compute the squared norm:
$$\|\mathbf{r}\|^2 = \|\hat{\mathbf{r}} + \mathbf{A}(\mathbf{x} - \hat{\mathbf{x}})\|^2$$
4. Expand using dot product properties:
$$\|\mathbf{r}\|^2 = \|\hat{\mathbf{r}}\|^2 + 2\hat{\mathbf{r}}^T\mathbf{A}(\mathbf{x} - \hat{\mathbf{x}}) + \|\mathbf{A}(\mathbf{x} - \hat{\mathbf{x}})\|^2$$
5. The cross term vanishes because $\hat{\mathbf{r}}$ is orthogonal to $R(\mathbf{A})$:
$$\hat{\mathbf{r}}^T\mathbf{A}(\mathbf{x} - \hat{\mathbf{x}}) = (\mathbf{A}^T\hat{\mathbf{r}})^T(\mathbf{x} - \hat{\mathbf{x}}) = \mathbf{0}^T(\mathbf{x} - \hat{\mathbf{x}}) = 0$$
6. Therefore:
$$\|\mathbf{r}\|^2 = \|\hat{\mathbf{r}}\|^2 + \|\mathbf{A}(\mathbf{x} - \hat{\mathbf{x}})\|^2 \geq \|\hat{\mathbf{r}}\|^2$$

This proves $\hat{\mathbf{x}}$ minimizes the residual!

5 Application: Linear Regression

5.1 Problem Setup

Scenario: We want to fit a straight line to data points (x_i, y_i), $i = 1, \ldots, m$.
Model: $y = \beta_0 + \beta_1 x$ (intercept β_0, slope β_1)
For each data point: $y_i \approx \beta_0 + \beta_1 x_i$

5.2 Matrix Formulation

We can write this as $\mathbf{A}\boldsymbol{\beta} \approx \mathbf{b}$:

$$\underbrace{\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}}_{\boldsymbol{\beta}} \approx \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}}_{\mathbf{b}}$$

5.3 Step-by-Step Example

Data: Points (1, 1), (2, 3), (3, 3), (4, 5)

1. **Construct matrices:**
$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 3 \\ 5 \end{bmatrix}$$
2. **Compute $\mathbf{A}^T\mathbf{A}$:**
$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}$$

3. **Compute $\mathbf{A}^T\mathbf{b}$:**

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 12 \\ 36 \end{bmatrix}$$

4. **Solve normal equations:**

$$\begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 12 \\ 36 \end{bmatrix}$$

5. **Solve the system:**

$$\begin{cases} 4\beta_0 + 10\beta_1 = 12 \\ 10\beta_0 + 30\beta_1 = 36 \end{cases}$$

   Multiply first equation by 2.5: $10\beta_0 + 25\beta_1 = 30$
   Subtract from second equation: $5\beta_1 = 6 \Rightarrow \beta_1 = 1.2$
   Substitute: $4\beta_0 + 10(1.2) = 12 \Rightarrow 4\beta_0 = 0 \Rightarrow \beta_0 = 0$

6. **Solution:** $\beta_0 = 0$, $\beta_1 = 1.2$
   Fitted line: $y = 1.2x$

7. **Verify:**

$$\mathbf{A}\hat{\boldsymbol{\beta}} = \begin{bmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{bmatrix}, \quad \hat{\mathbf{r}} = \mathbf{A}\hat{\boldsymbol{\beta}} - \mathbf{b} = \begin{bmatrix} 0.2 \\ -0.6 \\ 0.6 \\ -0.2 \end{bmatrix}$$

$$\|\hat{\mathbf{r}}\|^2 = 0.04 + 0.36 + 0.36 + 0.04 = 0.8$$

Matrix Method for Linear Regression

The linear regression model is:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where:

- $\mathbf{y}$ is the $n \times 1$ vector of responses
- $\mathbf{X}$ is the $n \times 2$ design matrix (first column is 1s for intercept)
- $\boldsymbol{\beta}$ is the 2×1 parameter vector $[\beta_0, \beta_1]^T$
- $\boldsymbol{\epsilon}$ is the $n \times 1$ error vector

The least squares solution is:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Example 1: Simple 2-Point Case

Data:

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad y = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Matrix Calculations:

1. **Step 1:** Construct design matrix $\mathbf{X}$:

$$\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

2. **Step 2:** Compute $\mathbf{X}^T\mathbf{X}$:

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}^T \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$$

3. **Step 3:** Compute $\mathbf{X}^T\mathbf{y}$:

$$\mathbf{X}^T\mathbf{y} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}^T \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

4. **Step 4:** Find $(\mathbf{X}^T\mathbf{X})^{-1}$:

$$\det(\mathbf{X}^T\mathbf{X}) = (2)(5) - (3)(3) = 10 - 9 = 1$$

$$(\mathbf{X}^T\mathbf{X})^{-1} = \frac{1}{1}\begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix}$$

5. **Step 5:** Compute $\hat{\boldsymbol{\beta}}$:

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 8 \end{bmatrix} = \begin{bmatrix} 25 - 24 \\ -15 + 16 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Result:

$$\beta_0 = 1, \quad \beta_1 = 1$$

Equation: $y = 1 + x$

Example 3: Medicine Dosage vs. Response Time

Data:

$$\text{Dosage (mg)} = \begin{bmatrix} 50 \\ 100 \\ 150 \\ 200 \\ 250 \end{bmatrix}, \quad \text{Response Time (min)} = \begin{bmatrix} 30 \\ 25 \\ 20 \\ 15 \\ 10 \end{bmatrix}$$

Matrix Solution:

1. **Step 1:** $\mathbf{X}$:

$$\mathbf{X} = \begin{bmatrix} 1 & 50 \\ 1 & 100 \\ 1 & 150 \\ 1 & 200 \\ 1 & 250 \end{bmatrix}$$

2. **Step 2:** $\mathbf{X}^T\mathbf{X}$:

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 5 & 750 \\ 750 & 137500 \end{bmatrix}$$

3. **Step 3:** $\mathbf{X}^T\mathbf{y}$:

$$\mathbf{X}^T\mathbf{y} = \begin{bmatrix} 100 \\ 12500 \end{bmatrix}$$

4. **Step 4:** $(\mathbf{X}^T\mathbf{X})^{-1}$:

$$\det(\mathbf{X}^T\mathbf{X}) = (5)(137500) - (750)(750) = 687500 - 562500 = 125000$$

$$(\mathbf{X}^T\mathbf{X})^{-1} = \frac{1}{125000}\begin{bmatrix} 137500 & -750 \\ -750 & 5 \end{bmatrix} = \begin{bmatrix} 1.1 & -0.006 \\ -0.006 & 0.00004 \end{bmatrix}$$

5. **Step 5:** $\hat{\boldsymbol{\beta}}$:

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} 1.1 & -0.006 \\ -0.006 & 0.00004 \end{bmatrix}\begin{bmatrix} 100 \\ 12500 \end{bmatrix} = \begin{bmatrix} 110 - 75 \\ -0.6 + 0.5 \end{bmatrix} = \begin{bmatrix} 35 \\ -0.1 \end{bmatrix}$$

Result:

$$\beta_0 = 35, \quad \beta_1 = -0.1$$

$$\text{Equation: Response Time} = 35 - 0.1 \times \text{Dosage}$$

Example 4: Body Temperature vs. Infection Score

Data:

$$\text{Temperature (°C)} = \begin{bmatrix} 37.0 \\ 37.5 \\ 38.0 \\ 38.5 \\ 39.0 \end{bmatrix}, \quad \text{Infection Score} = \begin{bmatrix} 1 \\ 3 \\ 5 \\ 7 \\ 9 \end{bmatrix}$$

Matrix Solution:

1. **Step 1:** $\mathbf{X}$:

$$\mathbf{X} = \begin{bmatrix} 1 & 37.0 \\ 1 & 37.5 \\ 1 & 38.0 \\ 1 & 38.5 \\ 1 & 39.0 \end{bmatrix}$$

2. **Step 2:** $\mathbf{X}^T\mathbf{X}$:

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 5 & 190 \\ 190 & 7222.5 \end{bmatrix}$$

3. **Step 3:** $\mathbf{X}^T\mathbf{y}$:

$$\mathbf{X}^T\mathbf{y} = \begin{bmatrix} 25 \\ 952.5 \end{bmatrix}$$

4. **Step 4:** $(\mathbf{X}^T\mathbf{X})^{-1}$:

$$\det(\mathbf{X}^T\mathbf{X}) = (5)(7222.5) - (190)(190) = 36112.5 - 36100 = 12.5$$

$$(\mathbf{X}^T\mathbf{X})^{-1} = \frac{1}{12.5}\begin{bmatrix} 7222.5 & -190 \\ -190 & 5 \end{bmatrix} = \begin{bmatrix} 577.8 & -15.2 \\ -15.2 & 0.4 \end{bmatrix}$$

5. **Step 5:** $\hat{\beta}$:

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} 577.8 & -15.2 \\ -15.2 & 0.4 \end{bmatrix}\begin{bmatrix} 25 \\ 952.5 \end{bmatrix} = \begin{bmatrix} 14445 - 14478 \\ -380 + 381 \end{bmatrix} = \begin{bmatrix} -33 \\ 1 \end{bmatrix}$$

Result:

$$\beta_0 = -33, \quad \beta_1 = 1$$

$$\text{Equation: Infection Score} = -33 + \text{Temperature}$$

Example 5: Cholesterol vs. Age (8 Patients)

Data:

$$\text{Age} = \begin{bmatrix} 30 \\ 35 \\ 40 \\ 45 \\ 50 \\ 55 \\ 60 \\ 65 \end{bmatrix}, \quad \text{Cholesterol} = \begin{bmatrix} 180 \\ 185 \\ 190 \\ 195 \\ 200 \\ 205 \\ 210 \\ 215 \end{bmatrix}$$

Matrix Solution:

1. **Step 1:** $\mathbf{X}$:

$$\mathbf{X} = \begin{bmatrix} 1 & 30 \\ 1 & 35 \\ 1 & 40 \\ 1 & 45 \\ 1 & 50 \\ 1 & 55 \\ 1 & 60 \\ 1 & 65 \end{bmatrix}$$

2. **Step 2:** $\mathbf{X}^T\mathbf{X}$:

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 8 & 380 \\ 380 & 19200 \end{bmatrix}$$

3. **Step 3:** $\mathbf{X}^T\mathbf{y}$:

$$\mathbf{X}^T\mathbf{y} = \begin{bmatrix} 1580 \\ 75800 \end{bmatrix}$$

4. **Step 4:** $(\mathbf{X}^T\mathbf{X})^{-1}$:

$$\det(\mathbf{X}^T\mathbf{X}) = (8)(19200) - (380)(380) = 153600 - 144400 = 9200$$

$$(\mathbf{X}^T\mathbf{X})^{-1} = \frac{1}{9200}\begin{bmatrix} 19200 & -380 \\ -380 & 8 \end{bmatrix} = \begin{bmatrix} 2.0870 & -0.0413 \\ -0.0413 & 0.00087 \end{bmatrix}$$

5. **Step 5:** $\hat{\beta}$:

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} 2.0870 & -0.0413 \\ -0.0413 & 0.00087 \end{bmatrix} \begin{bmatrix} 1580 \\ 75800 \end{bmatrix} = \begin{bmatrix} 3297.46 - 3130.54 \\ -65.254 + 65.946 \end{bmatrix} = \begin{bmatrix} 166.92 \\ 0.692 \end{bmatrix}$$

Result:

$$\beta_0 = 166.92, \quad \beta_1 = 0.692$$

$$\text{Equation: Cholesterol} = 166.92 + 0.692 \times \text{Age}$$

Example 6: Blood Sugar vs. Insulin Dose

Data:

$$\text{Insulin (units)} = \begin{bmatrix} 2 \\ 4 \\ 6 \\ 8 \\ 10 \end{bmatrix}, \quad \text{Blood Sugar (mg/dL)} = \begin{bmatrix} 200 \\ 180 \\ 160 \\ 140 \\ 120 \end{bmatrix}$$

Matrix Solution:

1. **Step 1:** $\mathbf{X}$:

$$\mathbf{X} = \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 6 \\ 1 & 8 \\ 1 & 10 \end{bmatrix}$$

2. **Step 2:** $\mathbf{X}^T\mathbf{X}$:

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 5 & 30 \\ 30 & 220 \end{bmatrix}$$

3. **Step 3:** $\mathbf{X}^T\mathbf{y}$:

$$\mathbf{X}^T\mathbf{y} = \begin{bmatrix} 800 \\ 4400 \end{bmatrix}$$

4. **Step 4:** $(\mathbf{X}^T\mathbf{X})^{-1}$:

$$\det(\mathbf{X}^T\mathbf{X}) = (5)(220) - (30)(30) = 1100 - 900 = 200$$

$$(\mathbf{X}^T\mathbf{X})^{-1} = \frac{1}{200} \begin{bmatrix} 220 & -30 \\ -30 & 5 \end{bmatrix} = \begin{bmatrix} 1.1 & -0.15 \\ -0.15 & 0.025 \end{bmatrix}$$

5. **Step 5:** $\hat{\beta}$:

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} 1.1 & -0.15 \\ -0.15 & 0.025 \end{bmatrix} \begin{bmatrix} 800 \\ 4400 \end{bmatrix} = \begin{bmatrix} 880 - 660 \\ -120 + 110 \end{bmatrix} = \begin{bmatrix} 220 \\ -10 \end{bmatrix}$$

Result:

$$\beta_0 = 220, \quad \beta_1 = -10$$

$$\text{Equation: Blood Sugar} = 220 - 10 \times \text{Insulin}$$

Matrix Operations Summary

Step-by-Step Algorithm:

1. Given data: $x = [x_1, x_2, \ldots, x_n]^T$, $y = [y_1, y_2, \ldots, y_n]^T$
2. Form $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ where $\mathbf{1}$ is $n \times 1$ vector of 1s
3. Compute $\mathbf{X}^T\mathbf{X}$
4. Compute $\mathbf{X}^T\mathbf{y}$
5. Compute $(\mathbf{X}^T\mathbf{X})^{-1}$
6. Compute $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$
7. Equation: $\hat{y} = \beta_0 + \beta_1 x$

Matrix Dimensions:

- $\mathbf{X}$: $n \times 2$
- $\mathbf{X}^T$: $2 \times n$
- $\mathbf{X}^T\mathbf{X}$: 2×2
- $\mathbf{X}^T\mathbf{y}$: 2×1
- $(\mathbf{X}^T\mathbf{X})^{-1}$: 2×2
- $\hat{\boldsymbol{\beta}}$: 2×1

Special Cases:

1. **Perfect Fit:** When residuals are zero
2. **Singular Matrix:** When $\det(\mathbf{X}^T\mathbf{X}) = 0$ (e.g., all x values are same)
3. **Underdetermined:** When $n < 2$ (not enough data)

Verification of Matrix Method

The matrix method is equivalent to solving:

$$\begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$$

This system comes from:

$$\frac{\partial}{\partial \beta_0} \sum (y_i - \beta_0 - \beta_1 x_i)^2 = 0$$

$$\frac{\partial}{\partial \beta_1} \sum (y_i - \beta_0 - \beta_1 x_i)^2 = 0$$

6 Practical Applications in Data Science & ML

6.1 Linear Regression (Foundation of ML)

- **Ordinary Least Squares (OLS):** Basic linear regression
- **Ridge Regression:** $\min \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + \lambda\|\mathbf{x}\|^2$ (adds L2 regularization)
- **Lasso Regression:** $\min \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + \lambda\|\mathbf{x}\|_1$ (adds L1 regularization, promotes sparsity)

6.2 Recommender Systems

- **Collaborative Filtering:** Predict user ratings for items
- **Matrix Factorization:** $\min \|\mathbf{R} - \mathbf{U}\mathbf{V}^T\|^2$ where $\mathbf{R}$ is rating matrix
- **Netflix Prize:** Famous competition won using matrix factorization techniques

6.3 Computer Vision

- **Image Reconstruction:** Reconstruct images from partial data
- **Structure from Motion:** Reconstruct 3D structure from 2D images
- **Camera Calibration:** Estimate camera parameters from known points

6.4 Natural Language Processing

- **Latent Semantic Analysis (LSA):** Reduce dimensionality of term-document matrices
- **Word Embeddings:** Methods like GloVe use matrix factorization

6.5 Signal Processing

- **Compressive Sensing:** Recover signals from fewer measurements
- **System Identification:** Estimate parameters of dynamical systems
- **Filter Design:** Design digital filters with desired frequency response

7 Numerical Considerations

7.1 Conditioning Issues

The matrix $\mathbf{A}^T\mathbf{A}$ can be **ill-conditioned** even when $\mathbf{A}$ is reasonably conditioned.
Condition number: $\kappa(\mathbf{A}^T\mathbf{A}) = [\kappa(\mathbf{A})]^2$
Example: If $\kappa(\mathbf{A}) = 10^3$, then $\kappa(\mathbf{A}^T\mathbf{A}) = 10^6$!

7.2 Better Numerical Methods

Instead of solving normal equations directly, use:

1. **QR Factorization:** $\mathbf{A} = \mathbf{Q}\mathbf{R}$, then solve $\mathbf{R}\mathbf{x} = \mathbf{Q}^T\mathbf{b}$
2. **Singular Value Decomposition (SVD):** $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, then $\hat{\mathbf{x}} = \mathbf{V}\mathbf{\Sigma}^{-1}\mathbf{U}^T\mathbf{b}$
3. **Iterative Methods:** For very large sparse systems (conjugate gradient)

8 Practice Problems

8.1 Problem 1: Basic Least Squares

Given the system:

$$\begin{cases} x_1 + x_2 = 2 \\ 2x_1 + x_2 = 3 \\ 3x_1 + x_2 = 5 \end{cases}$$

(a) Write in matrix form $\mathbf{A}\mathbf{x} = \mathbf{b}$
(b) Find the least squares solution
(c) Compute the residual norm

Solution 1

(a) Matrix form:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

(b) Compute normal equations:

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 14 & 6 \\ 6 & 3 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 23 \\ 10 \end{bmatrix}$$

Solve:

$$\begin{cases} 14x_1 + 6x_2 = 23 \\ 6x_1 + 3x_2 = 10 \end{cases}$$

Multiply second equation by 2: $12x_1 + 6x_2 = 20$
Subtract from first: $2x_1 = 3 \Rightarrow x_1 = 1.5$
Substitute: $6(1.5) + 3x_2 = 10 \Rightarrow 9 + 3x_2 = 10 \Rightarrow x_2 = \frac{1}{3}$
Solution: $x_1 = 1.5$, $x_2 = \frac{1}{3}$
(c) Residual:

$$\mathbf{r} = \mathbf{A}\mathbf{x} - \mathbf{b} = \begin{bmatrix} 1.5 + \frac{1}{3} - 2 \\ 3 + \frac{1}{3} - 3 \\ 4.5 + \frac{1}{3} - 5 \end{bmatrix} = \begin{bmatrix} -\frac{1}{6} \\ \frac{1}{3} \\ -\frac{1}{6} \end{bmatrix}$$

$$\|\mathbf{r}\|^2 = \frac{1}{36} + \frac{1}{9} + \frac{1}{36} = \frac{1}{36} + \frac{4}{36} + \frac{1}{36} = \frac{6}{36} = \frac{1}{6}$$

8.2 Problem 2: Polynomial Fitting

Fit a quadratic polynomial $y = a + bx + cx^2$ to the points: (-1, 2), (0, 1), (1, 3), (2, 5)

Solution 2

Model: $y = a + bx + cx^2$
For each point:

- (-1, 2): $a - b + c = 2$
- (0, 1): $a = 1$
- (1, 3): $a + b + c = 3$
- (2, 5): $a + 2b + 4c = 5$

Matrix form:

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ 3 \\ 5 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Normal equations:

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \\ 1 & 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 6 \\ 2 & 6 & 8 \\ 6 & 8 & 18 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \\ 1 & 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 11 \\ 11 \\ 31 \end{bmatrix}$$

Solve:

$$\begin{cases} 4a + 2b + 6c = 11 \\ 2a + 6b + 8c = 11 \\ 6a + 8b + 18c = 31 \end{cases}$$

Solving systematically gives: $a = -0.05$, $b = -1.15$, $c = 2.25$
Polynomial: $y = -0.05 - 1.15x + 2.25x^2$

8.3 Problem 3: Rank Deficiency

Consider:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

(a) What is the rank of $\mathbf{A}$?
(b) Show that the normal equations have infinitely many solutions
(c) Find the minimum residual norm

Solution 3

(a) Rank: All rows are identical, so rank$(\mathbf{A}) = 1$
(b) Normal equations:

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 & 9 \\ 6 & 12 & 18 \\ 9 & 18 & 27 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \\ 18 \end{bmatrix}$$

All equations are multiples of: $x_1 + 2x_2 + 3x_3 = 2$
This is a plane in $\mathbb{R}^3$ - infinitely many solutions!
(c) Minimum residual: Any solution satisfying $x_1 + 2x_2 + 3x_3 = 2$ gives:

$$\mathbf{A}\mathbf{x} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}, \quad \mathbf{r} = \mathbf{A}\mathbf{x} - \mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\|\mathbf{r}\|^2 = 1^2 + 0^2 + (-1)^2 = 2$$

All solutions give the same residual norm of 2.

9 Summary & Key Takeaways

9.1 Key Formulas

1. **Least Squares Problem:** $\min_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$
2. **Normal Equations:** $\mathbf{A}^T\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^T\mathbf{b}$
3. **Unique Solution:** $\hat{\mathbf{x}} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{b}$ (if $\mathbf{A}$ has full column rank)
4. **Residual:** $\hat{\mathbf{r}} = \mathbf{A}\hat{\mathbf{x}} - \mathbf{b}$

9.2 Geometric Insight

- Least squares finds the **orthogonal projection** of $\mathbf{b}$ onto $R(\mathbf{A})$
- The optimal residual is orthogonal to the column space
- The solution minimizes the Euclidean distance from $\mathbf{b}$ to $R(\mathbf{A})$

9.3 Practical Advice

1. **Check rank** of $\mathbf{A}$ before solving
2. **Avoid directly computing** $(\mathbf{A}^T\mathbf{A})^{-1}$ for numerical stability
3. Use **QR factorization** or **SVD** for better numerical solutions
4. Consider **regularization** (Ridge/Lasso) if $\mathbf{A}^T\mathbf{A}$ is ill-conditioned

9.4 Applications Checklist

- ✓ Linear regression and curve fitting
- ✓ System identification and parameter estimation
- ✓ Signal processing and filtering
- ✓ Machine learning (basis of many algorithms)
- ✓ Computer vision and image processing
- ✓ Recommender systems and collaborative filtering

10 Additional Resources

10.1 Textbooks:

- Strang, G. *Linear Algebra and Its Applications*
- Golub, G. H., & Van Loan, C. F. *Matrix Computations*
- Hastie, T., Tibshirani, R., & Friedman, J. *The Elements of Statistical Learning*

10.2 Online Courses:

- MIT OpenCourseWare: Linear Algebra (18.06)
- Coursera: Machine Learning by Andrew Ng
- edX: Linear Algebra - Foundations to Frontiers

10.3 Software Libraries:

- Python: `numpy.linalg.lstsq`, `scipy.linalg.lstsq`
- MATLAB: \ operator (backslash) or `mldivide`
- R: `lm()` function for linear models

Remember: Least squares is more than just a mathematical trick—it's a fundamental principle for extracting meaningful information from noisy data, with applications spanning science, engineering, and data analysis.